

Detailed Cleaned Dataset Summary Report

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Project Overview

The project focuses on healthcare recovery analytics and AI-based diet planning using structured patient monitoring datasets. The datasets were expanded and cleaned to support realistic healthcare analytics workflows.

Patients Dataset Description

The patients dataset contains demographic information, medical conditions, surgeries, medications, allergies, lifestyle details, smoking status, alcohol consumption, and registration details.

Daily Tracking Dataset Description

The tracking dataset contains time-series healthcare monitoring information including nutrition intake, meal compliance, hydration, sleep, exercise, blood pressure, blood sugar, symptoms, pain levels, and clinical notes.

Dataset Relationships

Both datasets are connected through `patient_id`, which acts as the primary relational key between master patient records and daily tracking records.

Cleaning Summary

The datasets underwent extensive preprocessing including missing value handling, datatype validation, duplicate removal, categorical normalization, blood pressure feature extraction, and realistic healthcare range validation.

Synthetic Data Generation

Synthetic healthcare records were generated using disease-aware logic, progressive recovery simulation, and doctor-style note generation to improve realism and support machine learning experimentation.

Healthcare Analytics Readiness

The cleaned datasets are suitable for exploratory data analysis, feature engineering, machine learning modeling, predictive analytics, and dashboard development.

Machine Learning Readiness

The final datasets are structured for classification, regression, clustering, recommendation systems, and healthcare recovery prediction workflows.

Conclusion

The cleaned and validated datasets provide a strong foundation for healthcare analytics and AI-driven recovery monitoring systems.